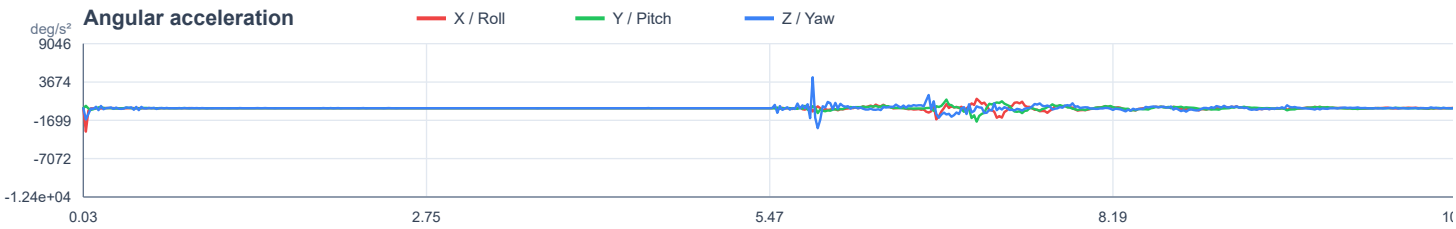
# Rocket Flight Log

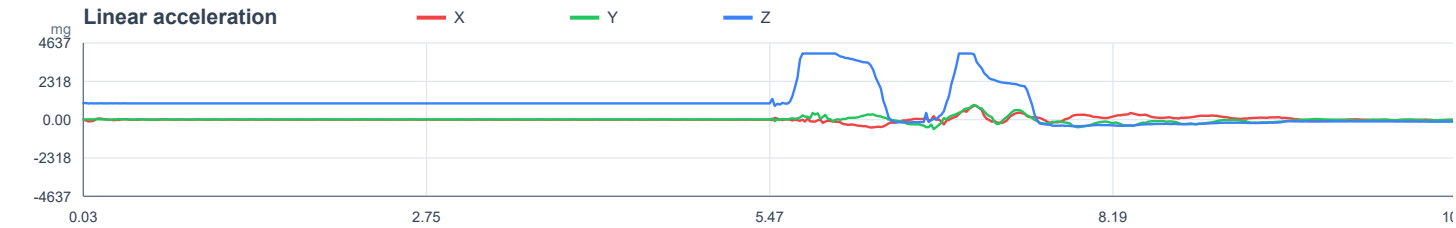
format v6 · 681 records · 50 Hz · 0.032–13.632 s

### Configuration snapshot

target_altitude_m: 80.0
stage_mass_kg: 0.4074
reference_diameter_m: 0.0616
air_density_kg_m3: 1.2
base_cd: 0.5
fin_drag_k: 0.02
fin_us_per_deg: 5.5556
servo_limit_deg: 10.0
roll_mix_sign: 1.0
pitch_mix_sign: -1.0
mount_roll_offset_deg: 180.0
target_roll_deg: 0.0
target_pitch_deg: 0.0
vertical_accel_sign: 1.0
ignition_delay_ms: 6700
ignition_pulse_ms: 800
force_deploy_time_ms: 13000
imu_period_ms: 10
barometer_period_ms: 40
rate_loop_period_ms: 10
angle_loop_period_ms: 20
full_flow_altitude_mix_enable: 1
servo_altitude_mix_enable: 0

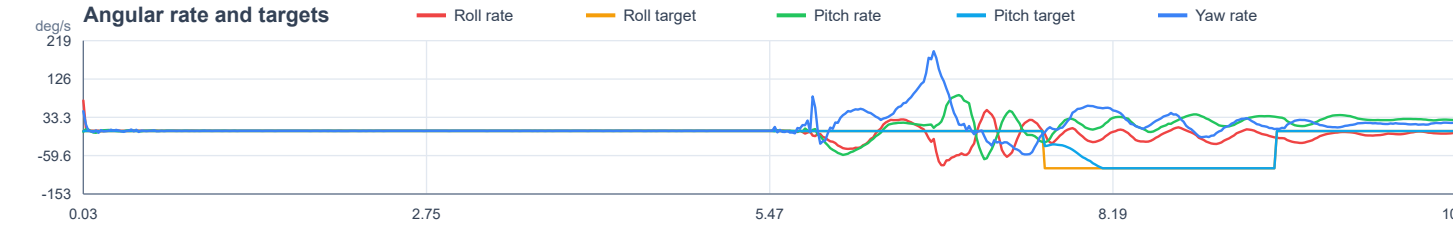
BMP address: 0x77 · IMU: ready · BMP init: ok

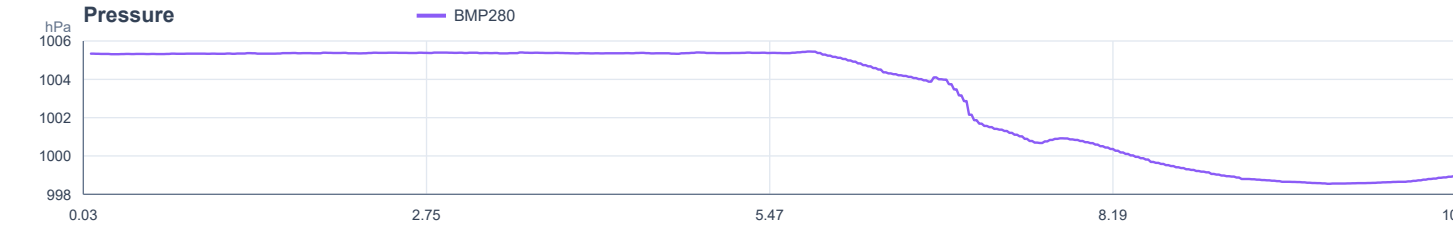


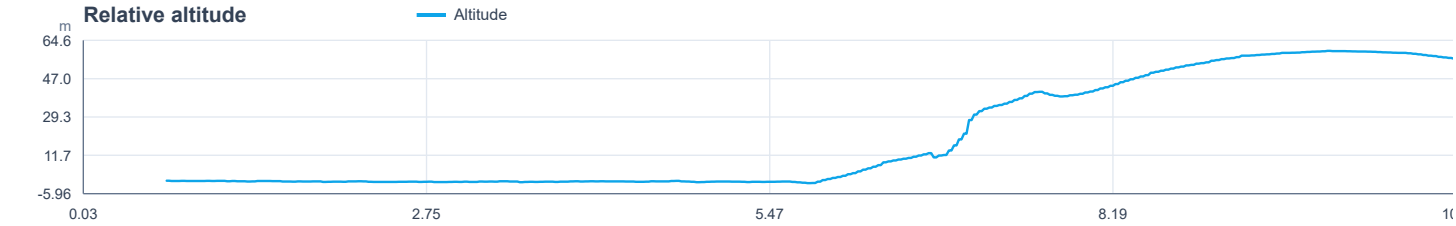



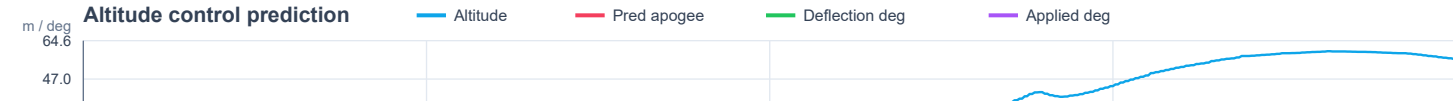


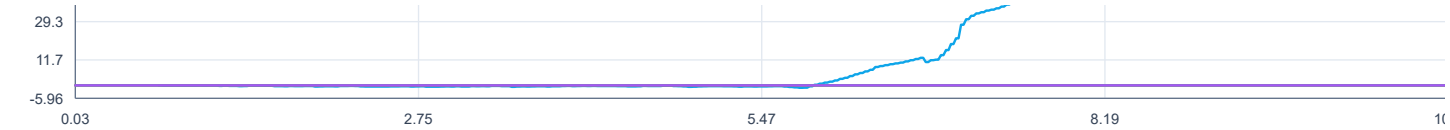




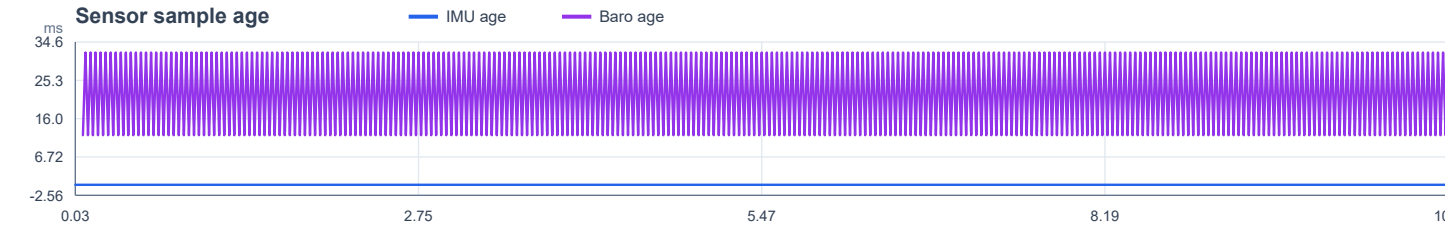



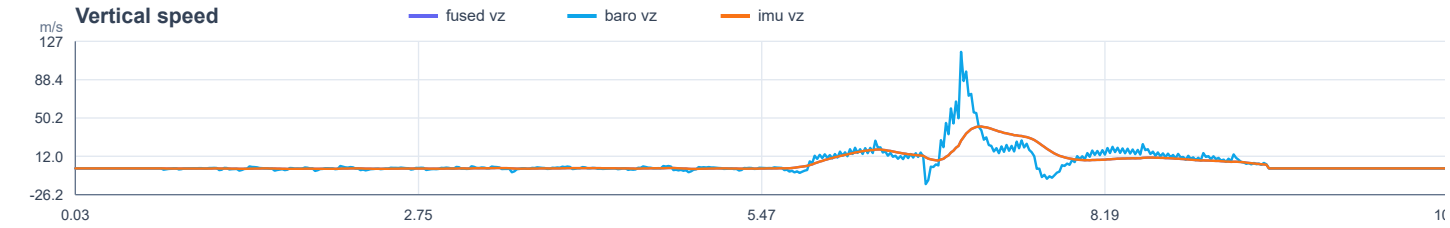



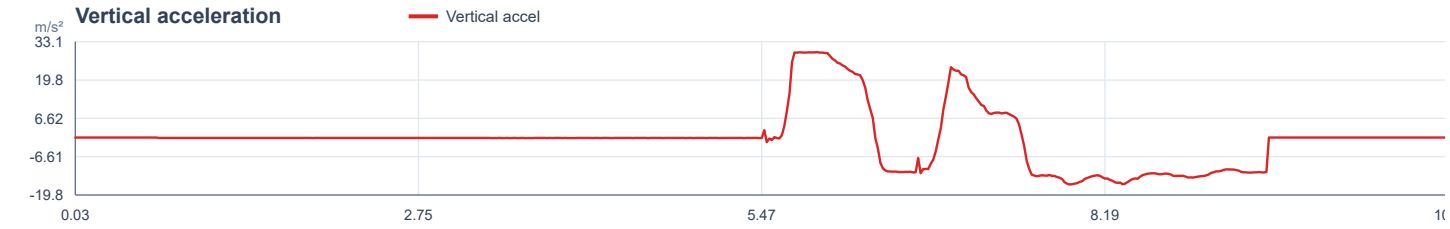

29.3
11.7
-5.96
0.03
2.75
5.47
8.19

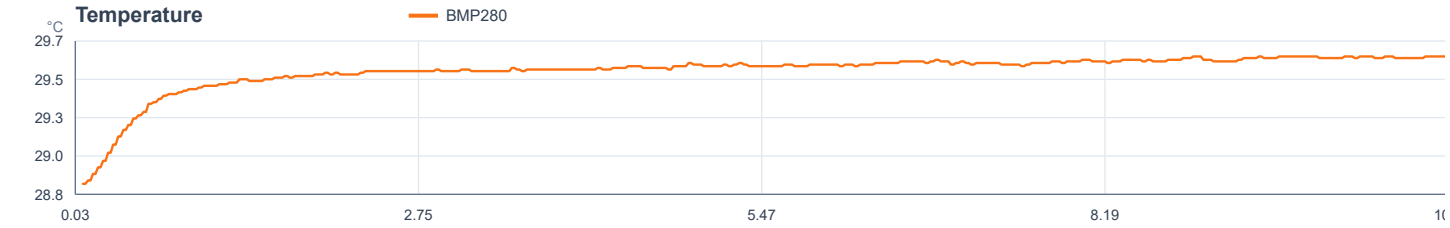

PID outputs
Roll PID
Pitch PID
us offset
10.4
-27.4
-65.3
-103
-141
0.03
2.75
5.47
8.19

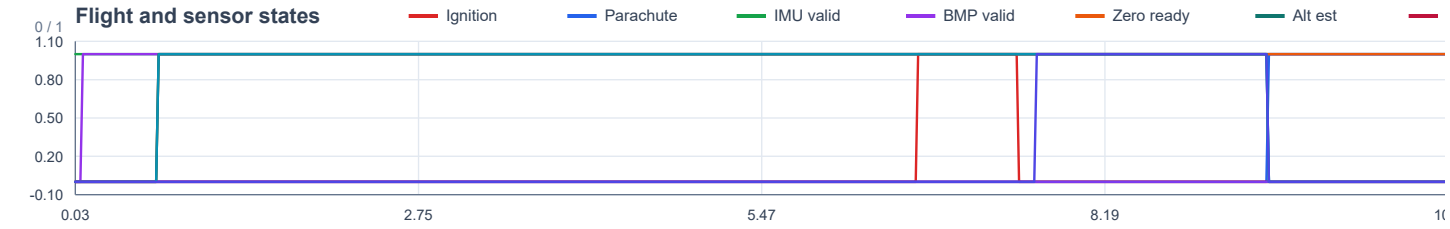

Actual servo PWM
Roll PWM
Pitch PWM
us
1649
1582
1516
1449
1382
0.03
2.75
5.47
8.19


Sensor sample age
IMU age
Baro age
ms
34.6
25.3
16.0
6.72
-2.56
0.03
2.75
5.47
8.19

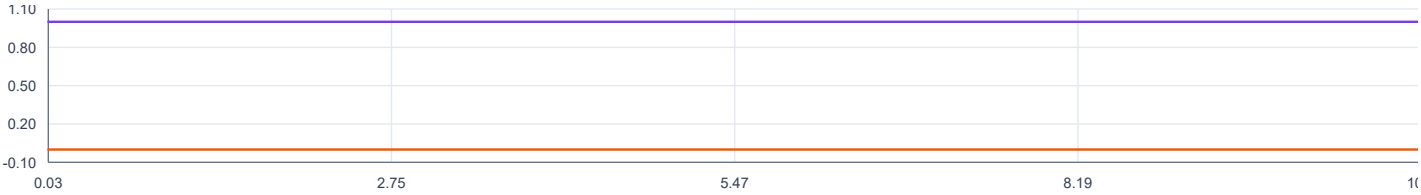

Vertical speed
fused vz
baro vz
imu vz
m/s
127
88.4
50.2
12.0
-26.2
0.03
2.75
5.47
8.19

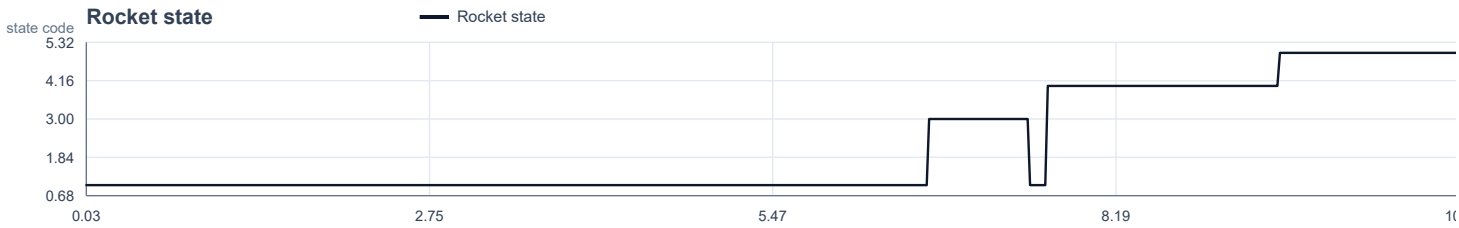

Vertical acceleration
Vertical accel
m/s²
33.1
19.8
6.62
-6.61
-19.8
0.03
2.75
5.47
8.19

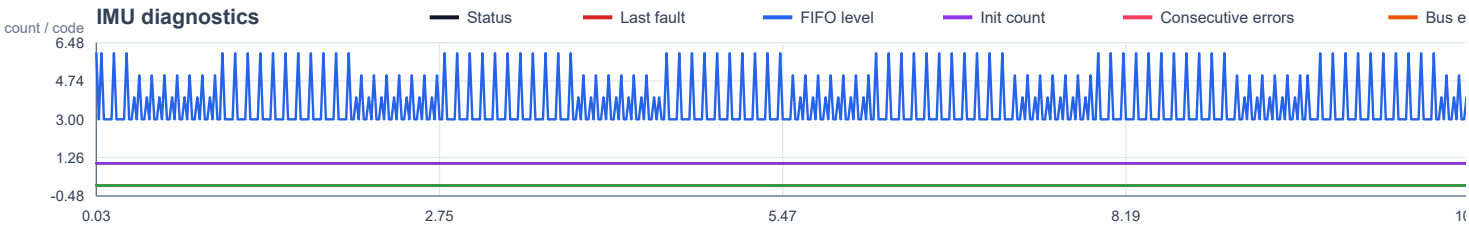

Temperature
BMP280
°C
29.7
29.5
29.3
29.0
28.8
0.03
2.75
5.47
8.19

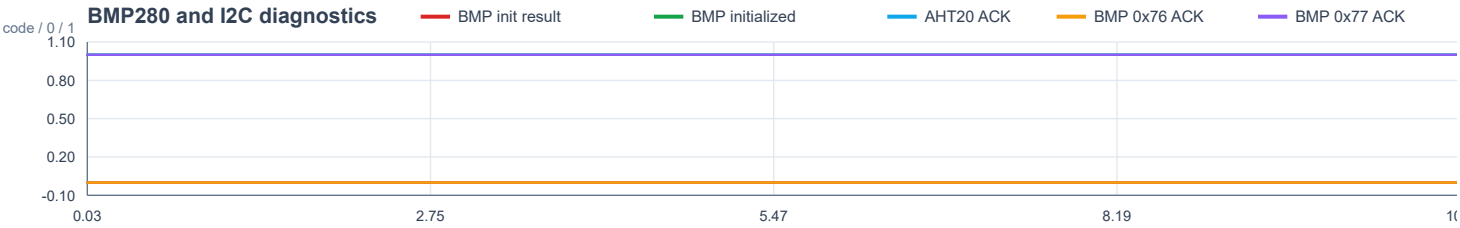

Flight and sensor states
Ignition
Parachute
IMU valid
BMP valid
Zero ready
Alt est
0 / 1
1.10
0.80
0.50
0.20
-0.10
0.03
2.75
5.47
8.19

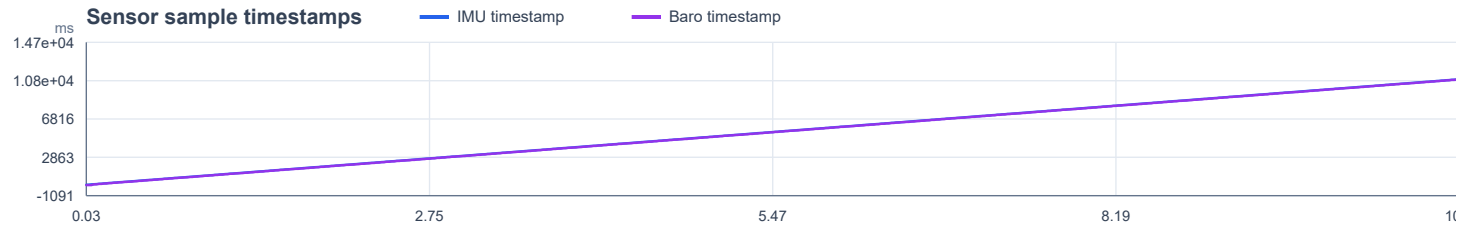

Control configuration states
Full-flow mix
Servo mix
0 / 1


1.10
0.80
0.50
0.20
-0.10
0.03
2.75
5.47
8.19


Rocket state
Rocket state
state code
5.32
4.16
3.00
1.84
0.68
0.03
2.75
5.47
8.19


IMU diagnostics
Status
Last fault
FIFO level
Init count
Consecutive errors
count / code
6.48
4.74
3.00
1.26
-0.48
0.03
2.75
5.47
8.19


BMP280 and I2C diagnostics
BMP init result
BMP initialized
AHT20 ACK
BMP 0x76 ACK
BMP 0x77 ACK
code / 0 / 1
1.10
0.80
0.50
0.20
-0.10
0.03
2.75
5.47
8.19


Sensor sample timestamps
IMU timestamp
Baro timestamp
ms
1.47e+04
1.08e+04
6816
2863
-1091
0.03
2.75
5.47
8.19